UC SAN DIEGO / ECE 251C

13

Vanishing moments and wavelet filter design

Executive summary

Vanishing moments produce small wavelet coefficients even when a signal is only approximately polynomial. We make this precise by subtracting a local Taylor polynomial and bounding the remainder, keeping track of every scale factor. We then translate the moment conditions into zeros of the wavelet filters. The resulting factorization viewpoint explains the Haar and Daubechies examples and shows how different allocations of the same product-filter zeros change phase and symmetry. The final example gives the biorthogonal 4.4 pair associated with JPEG 2000.

13.1 From exact cancellation to small coefficients

Lecture 12 showed that vanishing moments give exact cancellation when a signal agrees with a polynomial on the support of a wavelet. A signal encountered in an application need not be exactly polynomial, but it may admit an accurate local polynomial approximation. We now ask what is lost by replacing "polynomial" with "approximately polynomial." The approximation error will determine the size of the coefficients that remain after cancellation.

Recall the support–moment tradeoff for real, compactly supported, orthonormal dyadic MRA wavelets. A wavelet with p vanishing moments has support span at least $2p-1$, and its low-pass filter has length at least $2p$. Daubechies wavelets attain these bounds. Here filter length includes any zeros between the first and last nonzero taps. Thus greater cancellation has a cost in localization; the coefficient estimate will explain what is gained in return. The bounds are specific to the orthonormal setting and do not apply to arbitrary biorthogonal pairs.

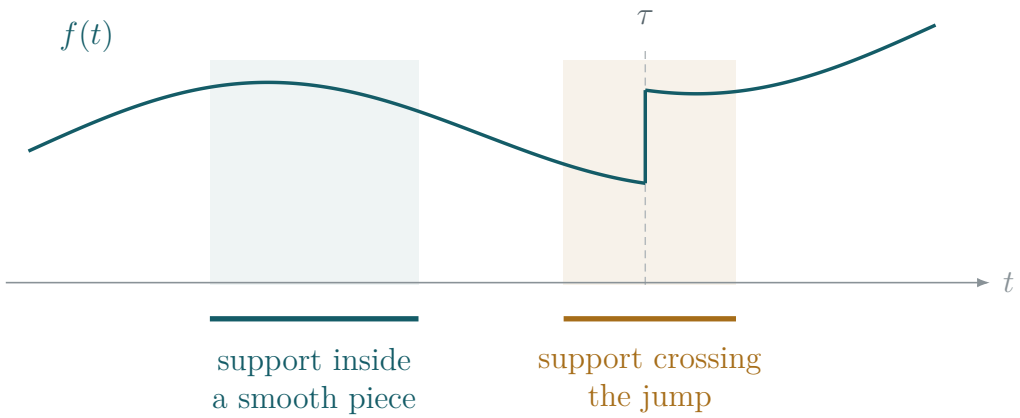


Figure 13.1: The estimate we will derive is local. It applies when the entire wavelet support, together with the Taylor expansion point, lies inside an interval of sufficient smoothness. Crossing a jump changes the problem.

13.2 The local polynomial model

Let $p \geq 1$ be an integer and let $\psi \in L^2(\mathbb{R})$ be compactly supported, with $\|\psi\|_{L^2} = 1$. Suppose that

$$\int_{\mathbb{R}} u^m \psi(u)\, \mathrm{d}u = 0, \qquad m = 0, \ldots, p-1. \tag{13.1}$$

As before, "p vanishing moments" means *at least* p. For a signal $f \in L^2(\mathbb{R})$, write

$$\psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n), \qquad d_i[n] = \int_{\mathbb{R}} f(t)\psi_{i,n}(t)^*\, \mathrm{d}t.$$

As throughout the notes, the inner product is linear in its first argument and conjugates the second. Fix $i, n \in \mathbb{Z}$ and set $t_0 = 2^{-i}n$. This is the reference point of the translated wavelet, not necessarily the center of its support. Suppose f has p continuous derivatives on an interval $\mathcal{I}$ containing both t_0 and $\operatorname{supp}\psi_{i,n}$, and that $|f^{(p)}(t)| \leq M$ throughout this interval.

The Taylor polynomial of degree at most $p-1$ is

$$T_{p-1}(t) = \sum_{m=0}^{p-1} \frac{f^{(m)}(t_0)}{m!}(t - t_0)^m. \tag{13.2}$$

For $p = 1$, it retains the local level; for $p = 2$, the level and slope; for $p = 3$, it also retains curvature. Define the remainder by

$$R_p(t) = f(t) - T_{p-1}(t).$$

Only this finite polynomial is needed. In particular, no analyticity or convergence of an infinite Taylor series is assumed. The derivative bound controls the error of the degree-$(p-1)$ approximation over the interval on which the wavelet coefficient is computed.

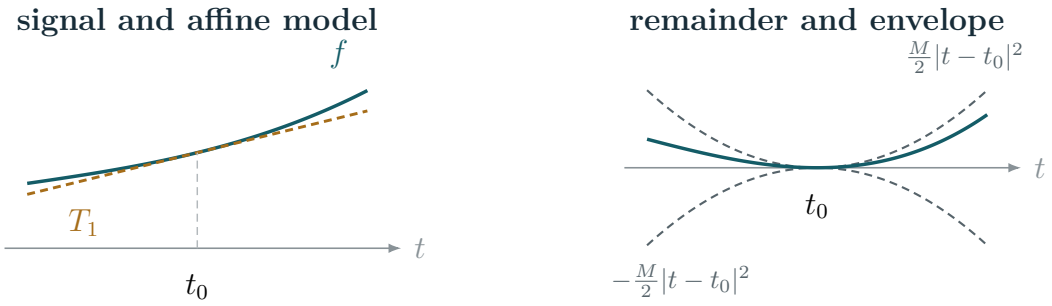


Figure 13.2: A local example with $v = t - t_0$: $f = 1 + 0.8v + v^2 + v^3$, $T_1 = 1 + 0.8v$, and $|v| \le 0.3$. Since $|f''| \le 3.8$, the signed remainder lies between $\pm(3.8/2)v^2$. The two panels use different vertical scales.

13.3 Bounding the Taylor remainder

Taylor's integral formula is

$$R_p(t) = \frac{1}{(p-1)!} \int_{t_0}^{t} (t-s)^{p-1} f^{(p)}(s)\, \mathrm{d}s. \tag{13.3}$$

For $p = 1$, this identity is the fundamental theorem of calculus. For larger p, integration by parts successively removes the Taylor terms. To make this explicit, denote the right-hand side by $Q_p(t)$. When $p \ge 2$, integration by parts gives

$$\begin{aligned} Q_p(t) &= \left[\frac{(t-s)^{p-1}}{(p-1)!} f^{(p-1)}(s) \right]_{t_0}^{t} + \frac{1}{(p-2)!} \int_{t_0}^{t} (t-s)^{p-2} f^{(p-1)}(s)\, \mathrm{d}s \\ &= Q_{p-1}(t) - \frac{f^{(p-1)}(t_0)}{(p-1)!} (t-t_0)^{p-1}. \end{aligned}$$

Starting from $Q_1(t) = f(t) - f(t_0)$, each step subtracts the next Taylor term. This proves (13.3).

If $t \ge t_0$, then

$$\begin{aligned} |R_p(t)| &\le \frac{1}{(p-1)!} \int_{t_0}^{t} (t-s)^{p-1} |f^{(p)}(s)|\, \mathrm{d}s \\ &\le \frac{M}{(p-1)!} \int_{t_0}^{t} (t-s)^{p-1}\, \mathrm{d}s = \frac{M}{p!} (t-t_0)^p. \end{aligned}$$

For $t < t_0$, reverse the integration limits before taking absolute values:

$$|R_p(t)| \le \frac{M}{(p-1)!} \int_{t}^{t_0} (s-t)^{p-1}\, \mathrm{d}s = \frac{M}{p!} (t_0-t)^p.$$

Thus both directions give the same bound:

$$\boxed{|f(t) - T_{p-1}(t)| \le \frac{M}{p!} |t-t_0|^p, \qquad t \in \mathcal{I}.} \tag{13.4}$$

The additional factor $1/p$ comes from integrating $(t-s)^{p-1}$. Together with $(p-1)!$, it gives $p!$. Thus, on an interval of width proportional to 2^{-i} around t_0, the Taylor error is bounded by a constant times $M2^{-ip}$. We next use vanishing moments to remove the Taylor polynomial itself.

13.4 The coefficient sees only the remainder

The moment conditions are stated for the mother wavelet, whereas the coefficient uses its translated and dilated copy. We therefore check the cancellation in the coordinates of that copy. For $0 \le m < p$, use $u = 2^i t - n$, so that

$$t - t_0 = 2^{-i}u, \qquad \mathrm{d}t = 2^{-i}\,\mathrm{d}u.$$

Then

$$\begin{aligned}\int_{\mathbb{R}} (t-t_0)^m \psi_{i,n}(t)^*\,\mathrm{d}t &= \int_{\mathbb{R}} (2^{-i}u)^m\, 2^{i/2}\psi(u)^*\, 2^{-i}\,\mathrm{d}u \\ &= 2^{-i(m+1/2)} \left(\int_{\mathbb{R}} u^m \psi(u)\,\mathrm{d}u\right)^* = 0.\end{aligned}$$

Every term of (13.2) is therefore canceled:

$$\int_{\mathbb{R}} T_{p-1}(t)\psi_{i,n}(t)^*\,\mathrm{d}t = 0.$$

These integrals involve only the compact wavelet support. We are not asserting that a polynomial on the whole line belongs to $L^2(\mathbb{R})$.

Now add and subtract the Taylor polynomial inside the coefficient:

$$\begin{aligned}d_i[n] &= \int_{\mathbb{R}} \big[f(t) - T_{p-1}(t) + T_{p-1}(t)\big]\psi_{i,n}(t)^*\,\mathrm{d}t \\ &= \int_{\operatorname{supp}\psi_{i,n}} \big[f(t) - T_{p-1}(t)\big]\psi_{i,n}(t)^*\,\mathrm{d}t.\end{aligned}$$

The triangle inequality and (13.4) imply

$$\begin{aligned}|d_i[n]| &\le \int_{\operatorname{supp}\psi_{i,n}} |f(t) - T_{p-1}(t)|\,|\psi_{i,n}(t)|\,\mathrm{d}t \\ &\le \frac{M}{p!}\int_{\mathbb{R}} |t-t_0|^p\, 2^{i/2}|\psi(2^i t - n)|\,\mathrm{d}t. \qquad (13.5)\end{aligned}$$

Extending the last integral to $\mathbb{R}$ changes nothing: the wavelet vanishes outside its support. Smoothness is needed only on $\mathcal{I}$.

> **Cancellation is automatic**
> The DWT does not fit and subtract a Taylor polynomial. The wavelet's moments make the polynomial's contribution zero without ever computing it.

13.5 Every factor in the scale-dependent estimate

In (13.5), make the increasing affine substitution

$$u = 2^i t - n, \qquad t = 2^{-i}(u+n), \qquad t_0 = 2^{-i} n.$$

Consequently,

$$|t - t_0|^p = 2^{-ip}|u|^p, \qquad \psi(2^i t - n) = \psi(u), \qquad \mathrm{d}t = 2^{-i}\,\mathrm{d}u.$$

The integration limits stay $-\infty$ and $+\infty$. Substitution gives

$$\begin{aligned} |d_i[n]| &\le \frac{M}{p!} \int_{\mathbb{R}} 2^{-ip}|u|^p \, 2^{i/2}|\psi(u)| \, 2^{-i}\,\mathrm{d}u \\ &= \frac{M}{p!}\, 2^{-ip+i/2-i} \int_{\mathbb{R}} |u|^p |\psi(u)|\,\mathrm{d}u \\ &= \frac{M}{p!}\, 2^{-i(p+1/2)} \int_{\mathbb{R}} |u|^p |\psi(u)|\,\mathrm{d}u. \end{aligned}$$

The powers of two come from three different sources: the degree-p remainder bound, the unit-energy amplitude of the wavelet, and the change of integration variable. Collecting them gives the exponent $p + 1/2$. The change of variables is exact; the inequality arose earlier when the remainder was replaced by its bound.

Theorem 13.1 (Local decay of wavelet coefficients)**.** Under the assumptions of Section 13.2,

$$\boxed{|d_i[n]| \le C_{\psi,p} M\, 2^{-i(p+1/2)}, \qquad C_{\psi,p} = \frac{1}{p!} \int_{\mathbb{R}} |u|^p |\psi(u)|\,\mathrm{d}u.} \tag{13.6}$$

The preceding calculation proves the theorem. Its constant is finite: on the compact support, $|u|^p$ is bounded, and Cauchy–Schwarz gives $\|\psi\|_{L^1} \le |\operatorname{supp}\psi|^{1/2}\|\psi\|_{L^2}$. The constant depends on ψ and p, but not on the scale or location. The local behavior of the signal enters through M; the factor $2^{-i(p+1/2)}$ records the effect of refining the scale.

$$\underbrace{2^{-ip}}_{\substack{\text{Taylor}\\ \text{error}}} \qquad \underbrace{2^{i/2}}_{\substack{\text{unit-energy}\\ \text{amplitude}}} \qquad \underbrace{2^{-i}}_{\substack{\text{integration}\\ \text{variable}}} = 2^{-i(p+1/2)}.$$

Increasing i shortens the analyzing interval. The order-p polynomial approximation improves by 2^{-p} per refinement; the remaining $2^{-1/2}$ comes from the coefficient normalization.

13.6 Rapid decay is not the same as an exact zero

Suppose the same derivative bound M holds on all the supports being considered. Then (13.6) gives a geometrically decreasing upper bound as the resolution index i increases. Equivalently, the decay is algebraic in the inverse support scale 2^i. The statement concerns an upper bound, not an equality or a monotone trajectory for each individual coefficient.

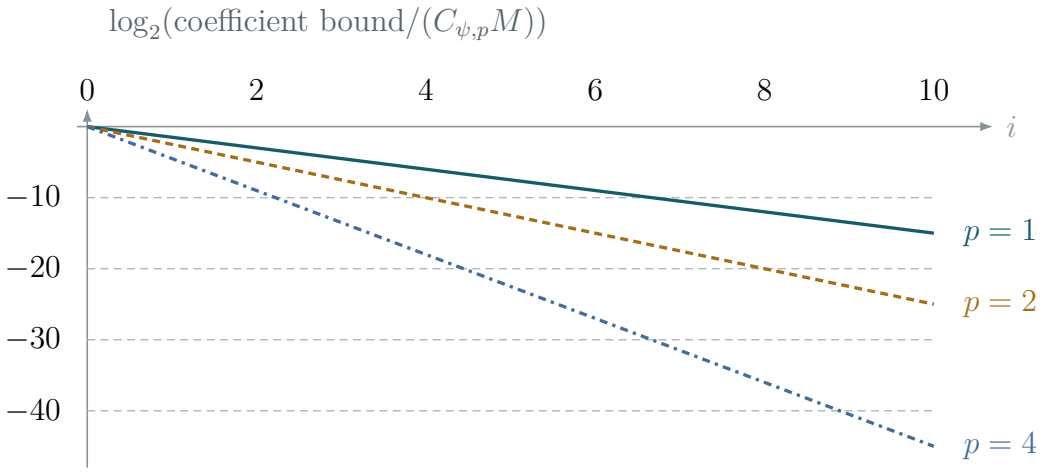


Figure 13.3: Normalized upper bounds, not measured coefficients. Their base-two logarithms have slope $-(p + 1/2)$. Different wavelets and derivative orders generally have different constants $C_{\psi,p}M$.

For example, the scale factor alone at $i = 10$ is

p	1	2	4
$2^{-10(p+1/2)}$	2^{-15}	2^{-25}	2^{-45}
	3.05×10^{-5}	2.98×10^{-8}	2.84×10^{-14}

These scale factors must be multiplied by $C_{\psi,p}M$. In particular, their size alone cannot justify calling a coefficient numerically zero. Multiplying the signal by a constant multiplies both its derivative bound and its coefficients by the same factor. A numerical decision must therefore include a specified tolerance.

For a tolerance $\varepsilon > 0$, a particular coefficient is guaranteed to satisfy $|d_i[n]| \le \varepsilon$ whenever

$$C_{\psi,p}M\, 2^{-i(p+1/2)} \le \varepsilon.$$

This connects the bound to deciding which details are negligible. It does not assert monotonicity of individual coefficients between scales, or supply the bound on supports crossing a singularity. A wavelet with p moments can use this argument with any smaller positive order for which the signal has the required derivative bound.

13.7 A worked example: Haar on a ramp

An affine signal separates exact cancellation from decay particularly clearly. Haar cancels its constant term but leaves a contribution from the slope. Computing that contribution will also check the exponent in the general estimate.

For Haar,

$$\psi(u) = \mathbf{1}_{[0,1/2)}(u) - \mathbf{1}_{[1/2,1)}(u), \qquad \int \psi(u)\,\mathrm{d}u = 0.$$

Its first moment is not zero:

$$\int_0^1 u\psi(u)\,\mathrm{d}u = \int_0^{1/2} u\,\mathrm{d}u - \int_{1/2}^1 u\,\mathrm{d}u = \frac{1}{8} - \frac{3}{8} = -\frac{1}{4}.$$

Suppose $f(t) = A + Bt$ on the wavelet support. The same substitution gives

$$\begin{aligned} d_i[n] &= 2^{-i/2}\int_0^1 \big[A + B2^{-i}(u+n)\big]\psi(u)\,\mathrm{d}u \\ &= 2^{-i/2}(A + B2^{-i}n)\underbrace{\int_0^1 \psi(u)\,\mathrm{d}u}_{0} + B2^{-3i/2}\underbrace{\int_0^1 u\psi(u)\,\mathrm{d}u}_{-1/4} \\ &= -\frac{B}{4}\,2^{-3i/2}. \end{aligned}$$

The constant part of the affine signal cancels, while the slope remains. Thus Haar produces nonzero detail coefficients on a ramp even though the ramp is smooth. Their magnitudes nevertheless decrease at the rate predicted for $p = 1$. In particular, taking $f(t) = t$ on the wavelet support gives $d_i[n] = -2^{-3i/2}/4$.

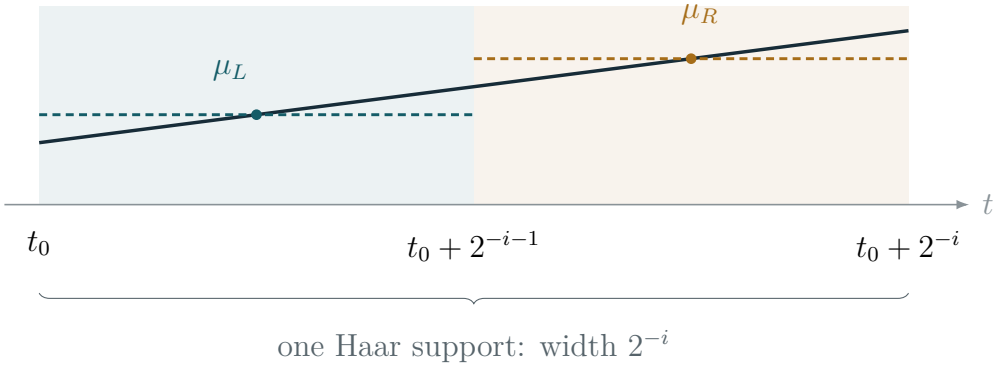


Figure 13.4: Haar compares the average over the left half of its support with the average over the right half. A positive slope makes the first average smaller, explaining the negative coefficient in our sign convention.

If these averages are μ_L, μ_R, each half-integral equals 2^{-i-1} times its average. Hence $d_i[n] = 2^{-i/2-1}(\mu_L - \mu_R)$. For the affine signal, $\mu_L - \mu_R = -B2^{-i-1}$, which gives the same result. A wavelet with two vanishing moments cancels the affine piece exactly.

13.8 The two-scale equations in the Fourier domain

The coefficient estimate explains the role of vanishing moments in approximation. To use it in filter design, we need an equivalent condition on the filter taps. We now derive that condition from the two-scale equations, so that the cancellation order can be prescribed before the scaling function and wavelet have been computed.

Assume now that ϕ, ψ are compactly supported functions from an orthonormal MRA, with $\int \phi(t)\,\mathrm{d}t = 1$, and FIR synthesis filters g_0, g_1. The two-scale equations are

$$\phi(t) = \sqrt{2}\sum_n g_0[n]\phi(2t-n), \qquad \psi(t) = \sqrt{2}\sum_n g_1[n]\phi(2t-n). \tag{13.7}$$

Recall our Fourier conventions:

$$\widehat{f}(\omega) = \int_{\mathbb{R}} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t, \qquad \widehat{g}_\ell(\omega) = \sum_n g_\ell[n]\mathrm{e}^{-\mathrm{j}\omega n} = G_\ell(\mathrm{e}^{\mathrm{j}\omega}).$$

For each term in (13.7), $u = 2t - n$ gives

$$\begin{aligned}\int_{\mathbb{R}} \sqrt{2}\,\phi(2t-n)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t &= \frac{1}{\sqrt{2}}\int_{\mathbb{R}} \phi(u)\mathrm{e}^{-\mathrm{j}(\omega/2)(u+n)}\,\mathrm{d}u \\ &= \frac{1}{\sqrt{2}}\mathrm{e}^{-\mathrm{j}\omega n/2}\widehat{\phi}(\omega/2).\end{aligned}$$

Summing and replacing ω by 2ω yields

$$\boxed{\widehat{\phi}(2\omega) = \frac{1}{\sqrt{2}}\widehat{g}_0(\omega)\widehat{\phi}(\omega), \qquad \widehat{\psi}(2\omega) = \frac{1}{\sqrt{2}}\widehat{g}_1(\omega)\widehat{\phi}(\omega).} \tag{13.8}$$

With $g_1[n] = (-1)^n g_0^*[K-n]$ and K odd, the conjugate-mirror identity is

$$\widehat{g}_1(\omega) = -\mathrm{e}^{-\mathrm{j}K\omega}\widehat{g}_0(\omega+\pi)^*.$$

Therefore

$$\widehat{\psi}(2\omega) = -\frac{\mathrm{e}^{-\mathrm{j}K\omega}}{\sqrt{2}}\widehat{g}_0(\omega+\pi)^*\widehat{\phi}(\omega). \tag{13.9}$$

The scaling-function factor is part of this identity and cannot be dropped. For the moment calculation, however, its important property is that it does not vanish near the origin: $\widehat{\phi}$ is continuous and $\widehat{\phi}(0) = 1$. Multiplication by this factor will therefore preserve the order of a zero.

13.9 Vanishing moments are spectral zeros

Compact support permits differentiation under the Fourier integral: every polynomial multiple of ψ is integrable. Thus

$$\widehat{\psi}'(\omega) = -\mathrm{j}\int_{\mathbb{R}} t\psi(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t,$$
$$\widehat{\psi}^{(m)}(\omega) = (-\mathrm{j})^m\int_{\mathbb{R}} t^m\psi(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t.$$

At zero frequency this becomes

$$\widehat{\psi}^{(m)}(0) = (-\mathrm{j})^m\int_{\mathbb{R}} t^m\psi(t)\,\mathrm{d}t. \tag{13.10}$$

The factor $(-\mathrm{j})^m$ is nonzero. Consequently, the moment of order m vanishes if and only if the derivative of order m of $\widehat{\psi}$ vanishes at the origin. The first p moment conditions therefore describe a zero of order at least p, with the value of the transform counted as the derivative of order zero.

A nonzero factor does not change this order. In (13.9), set $F = BC$, where

$$F(\omega) = \widehat{\psi}(2\omega), \quad B(\omega) = -\frac{\mathrm{e}^{-\mathrm{j}K\omega}}{\sqrt{2}}\widehat{\phi}(\omega), \quad C(\omega) = \widehat{g}_0(\omega+\pi)^*.$$

Here $B(0) = -1/\sqrt{2} \neq 0$. The product rule gives

$$F^{(m)}(0) = \sum_{r=0}^{m}\binom{m}{r}B^{(r)}(0)C^{(m-r)}(0).$$

If $C^{(r)}(0) = 0$ for $r < p$, then $F^{(m)}(0) = 0$ for $m < p$. Conversely, $F(0) = 0$ forces $C(0) = 0$. Inductively, after the lower derivatives of C vanish, the formula reduces to $F^{(m)}(0) = B(0)C^{(m)}(0)$, forcing the next derivative to vanish as well.

Finally, $F^{(m)}(0) = 2^m\widehat{\psi}^{(m)}(0)$ and $C^{(m)}(0) = \widehat{g}_0^{(m)}(\pi)^*$. We have proved

$$\boxed{\begin{array}{c}\psi \text{ has } p \text{ vanishing moments}\\ \iff \quad \widehat{g}_1^{(m)}(0) = 0 \quad (0 \le m < p)\\ \iff \quad \widehat{g}_0^{(m)}(\pi) = 0 \quad (0 \le m < p).\end{array}} \tag{13.11}$$

The high-pass version follows from the second identity in (13.8) by the same nonzero-factor argument. This proof allows complex filters.

13.10 From frequency derivatives to factors of the filter

Differentiating the finite DTFT sum gives

$$\widehat{g}_0^{(m)}(\pi) = (-\mathrm{j})^m\sum_n(-1)^n n^m g_0[n].$$

Hence the moment conditions can be checked directly on the low-pass taps:

$$\boxed{\sum_n (-1)^n n^m g_0[n] = 0, \qquad m = 0, \dots, p-1.} \tag{13.12}$$

For $m = 0$, take $n^0 = 1$ at every index, including $n = 0$. This recovers the alternating-sum condition from Lecture 10. Each additional moment condition forces another derivative of the low-pass response to vanish at π. The conditions are linear in the taps, although orthogonality imposes additional quadratic constraints.

For a filter supported in $\{0, \dots, K\}$, set $P(v) = \sum_{n=0}^{K} g_0[n] v^n$, so $G_0(z) = P(z^{-1})$. If $P(v) = (v+1)^r Q(v)$ with $Q(-1) \neq 0$, then

$$P(\mathrm{e}^{-\mathrm{j}(\pi+\eta)}) = (1 - \mathrm{e}^{-\mathrm{j}\eta})^r Q(-\mathrm{e}^{-\mathrm{j}\eta}).$$

Since $1 - \mathrm{e}^{-\mathrm{j}\eta} = \mathrm{j}\eta + O(\eta^2)$, this has a zero of order exactly r at $\eta = 0$. Thus (13.11) is equivalent to

$$\boxed{G_0(z) = (1 + z^{-1})^p Q_0(z),} \tag{13.13}$$

where Q_0 is a polynomial in z^{-1}. Equivalently, G_0 has a zero at $z = -1$ of multiplicity at least p. The high-pass filter has the corresponding factor $(1 - z^{-1})^p$ at $z = 1$.

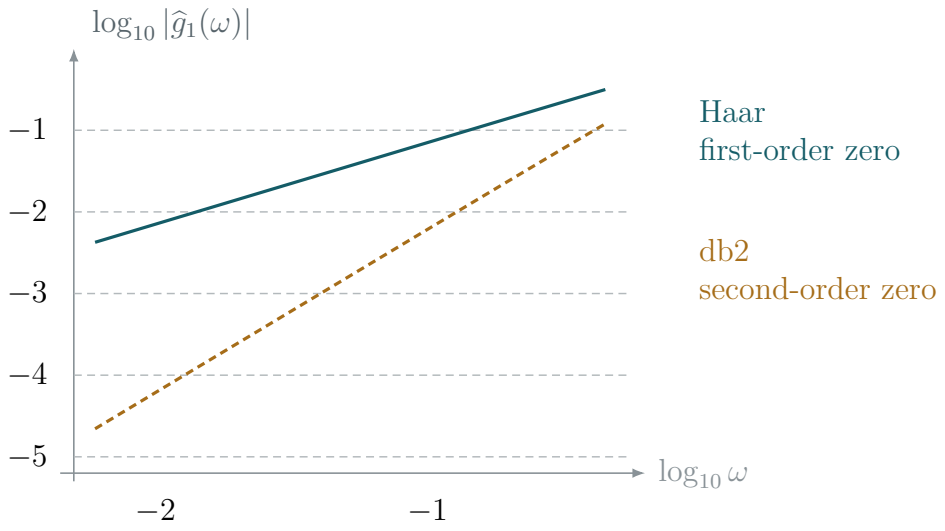


Figure 13.5: Low-frequency high-pass magnitudes for Haar and db2. A logarithmic plot reveals first- and second-order zeros: the responses behave as constant multiples of $|\omega|$ and $|\omega|^2$ near zero.

13.11 Product filters and the causal indexing convention

We now return to product-filter design. The examples use the causal indexing convention of Lecture 9, rather than the zero-delay analysis convention used in the

wavelet derivations. We describe the conversion explicitly so that the change of phase and reconstruction delay can be tracked. For synthesis taps $g_0[0], \dots, g_0[K]$, the zero-delay analysis transform is

$$H_0^{\text{zero}}(z) = [G_0(1/z^*)]^* .$$

Delaying it by K makes it causal. In the examples below, the symbol H_0 denotes this delayed filter:

$$H_0(z) = z^{-K} [G_0(1/z^*)]^* . \tag{13.14}$$

For real taps this is $H_0(z) = z^{-K} G_0(z^{-1})$, or $h_0[n] = g_0[K-n]$. This delay changes phase and indexing, but not magnitudes or vanishing-moment orders.

Define $P_0(z) = H_0(z)G_0(z)$. In the orthogonal case,

$$P_0(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-\mathrm{j}K\omega} |G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 .$$

Power complementarity and odd K give

$$\boxed{P_0(z) - P_0(-z) = 2z^{-K}.} \tag{13.15}$$

For FIR filters, equality on the unit circle is also equality of the Laurent polynomials: multiply by a sufficiently large power of z, then use that a nonzero polynomial cannot have infinitely many roots.

More generally, any FIR factorization $P_0 = H_0 G_0$ satisfying (13.15) gives a perfect-reconstruction bank by setting

$$H_1(z) = -G_0(-z), \qquad G_1(z) = H_0(-z).$$

Indeed,

$$\begin{aligned} G_0(z)H_0(-z) + G_1(z)H_1(-z) &= 0, \\ G_0(z)H_0(z) + G_1(z)H_1(z) &= P_0(z) - P_0(-z) = 2z^{-K}. \end{aligned}$$

The input–output relation is therefore $Y(z) = z^{-K} X(z)$. Both high-pass signs may be reversed together without changing reconstruction. Factorization alone guarantees this reconstruction identity, not orthogonality.

13.12 Haar: divide two zeros into one and one

Start with the three-tap product filter

$$P_0(z) = \frac{1}{2}(1 + z^{-1})^2 = \frac{1}{2} + z^{-1} + \frac{1}{2} z^{-2}. \tag{13.16}$$

Its odd coefficient is the unit tap at index one, so

$$P_0(z) - P_0(-z) = 2z^{-1}.$$

A balanced factorization is

$$G_0(z) = H_0(z) = \frac{1}{\sqrt{2}}(1 + z^{-1}).$$

Each factor has DC gain $\sqrt{2}$ and one zero at $z = -1$. The high-pass formulas from the preceding section give

$$G_1(z) = \frac{1}{\sqrt{2}}(1 - z^{-1}), \qquad H_1(z) = -\frac{1}{\sqrt{2}}(1 - z^{-1}).$$

Thus the complete causal bank reconstructs with a one-sample delay. This is the same Haar transform as before, with the analysis filters delayed.

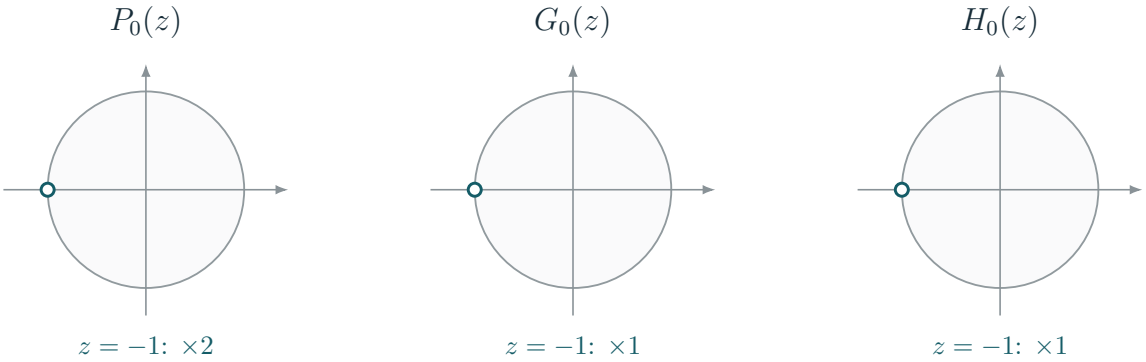


Figure 13.6: Factorization of the Haar product: $P_0 = G_0 H_0$. Open circles mark zeros, and $\times r$ below a panel gives the multiplicity at $z = -1$. The gray circle is $|z| = 1$; the horizontal and vertical axes are $\mathrm{Re}\, z$ and $\mathrm{Im}\, z$, with equal scales. Poles at the origin are omitted throughout these zero diagrams.

The product has two zeros at -1, but each wavelet low-pass factor has only one. Consequently, Haar has one vanishing moment, not two. In the orthogonal construction, the product's multiplicity is twice the wavelet's moment order.

The alternative names are db1 and D2: the first counts vanishing moments, while the second counts the two filter taps. We use this distinction again for the longer Daubechies filters.

13.13 The four-tap Daubechies factorization

The next example is the four-tap Daubechies pair from Homework 3. Its product filter has four zeros at -1, so an orthogonal factorization will give two to each low-pass factor. Consider the seven-tap product

$$P_0(z) = \frac{1}{16}\big(-1 + 9z^{-2} + 16z^{-3} + 9z^{-4} - z^{-6}\big). \tag{13.17}$$

Only the central coefficient has an odd index, and it equals one. Therefore $P_0(z) - P_0(-z) = 2z^{-3}$. Factoring gives

$$P_0(z) = \frac{1}{16}(1 + z^{-1})^4(-1 + 4z^{-1} - z^{-2}). \tag{13.18}$$

For example, multiplying the coefficient lists $(1, 4, 6, 4, 1)$ and $(-1, 4, -1)$ produces $(-1, 0, 9, 16, 9, 0, -1)$, verifying the factorization.

Besides four zeros at -1, the product has the two roots of

$$-z^2 + 4z - 1 = 0, \qquad \alpha = 2 - \sqrt{3}, \quad \beta = 2 + \sqrt{3}, \quad \alpha\beta = 1.$$

Assign two zeros at -1 and the inside root α to G_0. Its causal reversed partner H_0 receives the outside root β. Normalizing both DC gains to $\sqrt{2}$ gives

$$G_0(z) = \frac{(1 + z^{-1})^2}{4\sqrt{2}}\big[(1 + \sqrt{3}) + (1 - \sqrt{3})z^{-1}\big], \tag{13.19}$$

$$H_0(z) = \frac{(1 + z^{-1})^2}{4\sqrt{2}}\big[(1 - \sqrt{3}) + (1 + \sqrt{3})z^{-1}\big].$$

Their residual linear factors multiply to $2(-1 + 4z^{-1} - z^{-2})$; their common denominator is 32. Their product is therefore exactly (13.18).

Expanding G_0 recovers

$$\boxed{g_0 = \frac{1}{4\sqrt{2}}(1 + \sqrt{3},\ 3 + \sqrt{3},\ 3 - \sqrt{3},\ 1 - \sqrt{3}).} \tag{13.20}$$

Here h_0 is the reversed list. Each low-pass factor has exactly two zeros at -1, so the associated orthogonal wavelet has exactly two vanishing moments. This is db2, also called D4.

The db2 zero allocation. The inside and outside residual roots may be exchanged between the two factors. This reverses the tap lists, preserving their magnitude responses and moment orders while changing their phases. Thus the product determines the power response without uniquely fixing the individual filters.

Why a linear-phase product need not have linear-phase factors. A real symmetric FIR sequence $a[0], \ldots, a[N]$ satisfies $a[n] = a[N - n]$. Its response can be written

$$A(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-\mathrm{j}N\omega/2} \sum_{n=0}^{N} a[n]\mathrm{e}^{-\mathrm{j}\omega(n - N/2)}.$$

Pairing terms at n and $N - n$ turns the sum into real cosines. The response therefore has linear phase apart from π changes where this real amplitude changes sign.

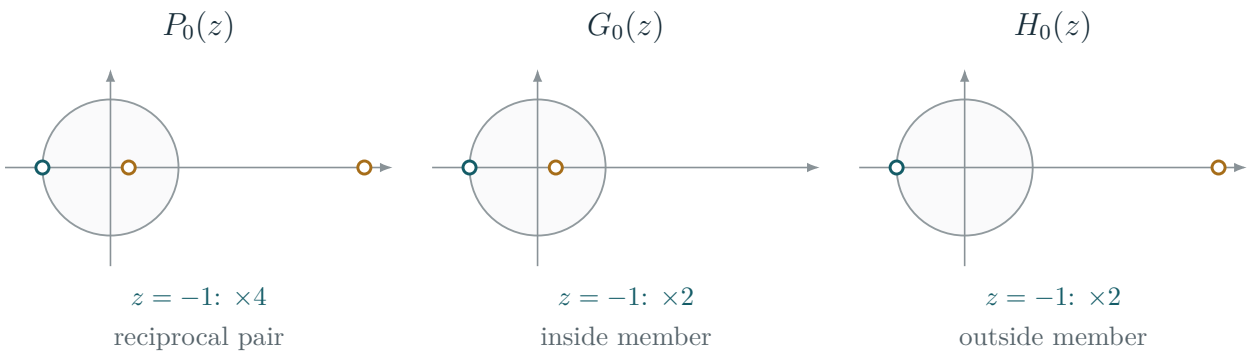


Figure 13.7: The four zeros at -1 are split equally. The remaining reciprocal pair is split between the two filters: $\alpha \approx 0.268$ and $\beta \approx 3.732$. All three panels use the same scale.

The symmetric seven-tap product in (13.17) has Type-I linear phase and delay three. The four-tap factors are neither symmetric nor antisymmetric, so their phases are nonlinear. Haar's symmetric two-tap factors are Type II.

Minimum- and maximum-phase terminology. After removing $(1 + z^{-1})^p$, the residual zeros of G_0 lie inside the unit circle and those of H_0 lie outside. These are the usual minimum- and maximum-phase spectral-factor choices. The full low-pass filters still have their required zeros at -1, so their inverses have poles on the unit circle. In particular, this terminology does not assert that the full filters possess stable causal inverses.

13.14 Four moments: the product behind the db4 diagram

The four-tap example has only one residual reciprocal pair to allocate. The eight-tap Daubechies example has both real and complex residual roots, so it also shows why conjugate roots must be kept together. Here $p = 4$ and $K = 7$. A concrete product underlying its zero diagram is

$$P_0(z) = z^{-7} + \frac{1}{2048}\big[-5(1 + z^{-14}) + 49(z^{-2} + z^{-12}) - 245(z^{-4} + z^{-10}) + 1225(z^{-6} + z^{-8})\big]. \tag{13.21}$$

Every term except z^{-7} has even exponent, so $P_0(z) - P_0(-z) = 2z^{-7}$ immediately. To read its zeros, write $v = z^{-1}$ and define

$$Q_4(v) = -5 + 40v - 131v^2 + 208v^3 - 131v^4 + 40v^5 - 5v^6.$$

Direct multiplication gives

$$\boxed{P_0(z) = \frac{1}{2048}(1 + z^{-1})^8 Q_4(z^{-1}).} \tag{13.22}$$

The factorization separates the roots fixed by the moment conditions from those available for phase design. Eight roots are at -1; the other six must be allocated between the low-pass factors while respecting the requirement that their coefficients be real.

Why the remaining roots occur in reciprocal-conjugate groups. The polynomial Q_4 has real, symmetric coefficients, so

$$Q_4(v)^* = Q_4(v^*), \qquad Q_4(v) = v^6 Q_4(v^{-1}).$$

Thus a nonzero root brings its conjugate and reciprocal as roots. Numerically, the three inside roots in the z-plane are

$$r_1 \approx 0.328875918, \qquad r_\pm \approx 0.284096298 \ \pm \ 0.243228226\mathrm{j}.$$

The outside roots are their reciprocals (the conjugate pair is unchanged as a set by reciprocal conjugation). All are away from the unit circle.

Choose the inside roots and four copies of -1 for G_0, and choose the outside roots and four copies of -1 for H_0. Each filter then has degree seven in z^{-1}, hence eight taps. Choosing conjugate roots together ensures real coefficients.

The db4 spectral factorization. If $\mathcal{R} = \{r_1, r_+, r_-\}$ is the inside root set, then

$$G_0(z) = c(1 + z^{-1})^4 \prod_{r \in \mathcal{R}} (1 - r z^{-1}), \qquad c = \frac{\sqrt{2}}{16 \prod_{r \in \mathcal{R}} (1 - r)}.$$

This expression fixes the gain by $G_0(1) = \sqrt{2}$. Define $H_0(z) = z^{-7} G_0(z^{-1})$. The product has exactly the zeros of (13.22); its DC gain is two, equal to $P_0(1)$. Therefore it equals P_0 identically.

The product identity and the reversal relation imply

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 + |G_0(\mathrm{e}^{\mathrm{j}(\omega+\pi)})|^2 = 2.$$

Together with $G_0(1) = \sqrt{2}$, this gives the orthogonal low-pass conditions. Its only unit-circle zero is -1, so the nonvanishing condition on $[-\pi/2, \pi/2]$ from Lecture 11 also holds. Thus this filter gives the orthonormal scaling function and wavelet described there. The order-four zero at -1 gives four vanishing moments.

The product has Type-I linear phase, but the individual eight-tap filters do not. The distinction is the same as for db2: symmetry of a product does not force symmetry of its factors. The remaining reciprocal-root choices change phase without changing the prescribed magnitude or cancellation order.

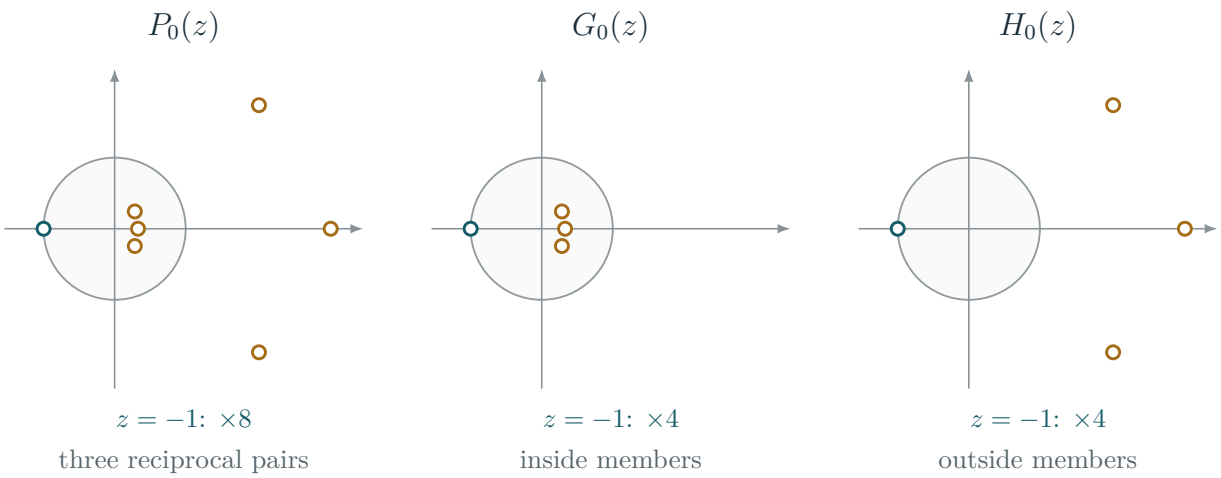


Figure 13.8: For db4 (D8), the product has eight zeros at -1. Each low-pass factor receives four. Splitting every residual reciprocal pair between the factors gives an orthogonal spectral factorization.

13.15 Keep reciprocal pairs together to obtain symmetry

The same product admits a different allocation. Again give four zeros at -1 to each low-pass factor, but place the entire complex reciprocal-conjugate quartet in G_0 and the real reciprocal pair in H_0. Keeping reciprocal roots together, rather than splitting each pair between reversed factors, allows both filters to be symmetric.

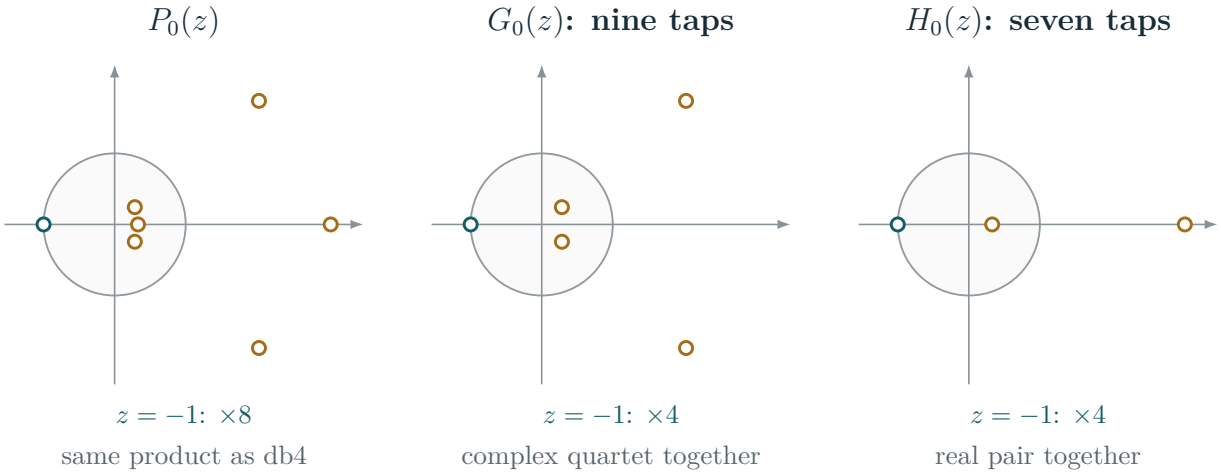


Figure 13.9: The biorthogonal 4.4 allocation keeps reciprocal pairs within a factor instead of splitting them between factors. The low-pass factors now have nine and seven taps, and each is symmetric. The product is unchanged.

Count the taps. The nine-tap factor has $4 + 4 = 8$ zeros, hence degree eight. The seven-tap factor has $4 + 2 = 6$ zeros, hence degree six. Both have four zeros at -1.

Check symmetry algebraically. A real reciprocal pair contributes

$$(1 - rz^{-1})(1 - r^{-1}z^{-1}) = 1 - (r + r^{-1})z^{-1} + z^{-2},$$

which has symmetric coefficients. A complex quartet gives the product of two such reciprocal-pair factors; including the conjugate pair makes the coefficients real. The factor $(1 + z^{-1})^4$ is also symmetric. Products of symmetric coefficient lists remain symmetric, because reversing a convolution reverses both factors. Thus both resulting filters are Type-I linear phase.

Normalize their DC gains to $\sqrt{2}$. Their product has exactly the same roots, degree, and DC gain as P_0, so it equals P_0. Perfect reconstruction still follows from (13.15), now with delay seven. The filters are no longer a reversed orthogonal pair.

A biorthogonal preview: the CDF 9/7 pair. This root allocation produces the standard biorthogonal 4.4 / CDF 9/7 low-pass pair, up to exchanging the two factors, delay choices, and normalization. Our figure labels follow the handwritten allocation: G_0 has nine taps and H_0 has seven.

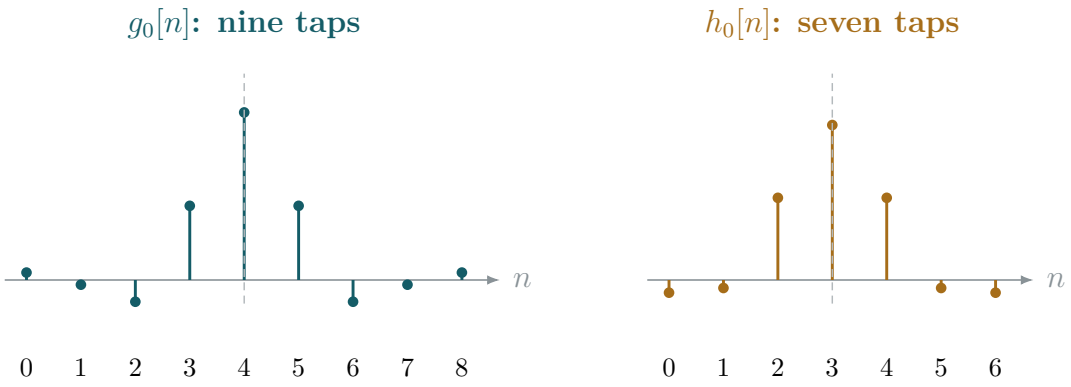


Figure 13.10: Numerical taps from the root allocation in the preceding construction, with both sums normalized to $\sqrt{2}$. The centers of symmetry are four and three. Internal zeros have not been inserted to make the lists artificially equal in length.

The nine-tap analysis low-pass and seven-tap synthesis low-pass are used in the usual orientation of JPEG 2000's irreversible CDF 9/7 transform.[1] The notation 4.4 refers to four vanishing moments on each of the two wavelet sides, whereas 9/7 records the two filter lengths. This is not the same wavelet family as db4, despite the shared product polynomial.

In a biorthogonal representation, the synthesis and analysis functions need not coincide. Instead, two families satisfy $\langle q_\lambda, \tilde{q}_\mu \rangle = \delta_{\lambda\mu}$. Taking the inner product

[1] This application is specified in ITU-T T.800 (2002), Annex F and Tables J.1–J.2. The standard uses a different gain normalization. Interchanging the two low-pass factors gives the dual orientation shown here.

of $f = \sum_\lambda c_\lambda q_\lambda$ with $\tilde{q}_\mu$ recovers the coefficient c_μ. It is therefore the moments of the analyzing wavelet that determine polynomial cancellation. The earlier decay argument uses these moments, not orthogonality; a different normalization changes the constant $C_{\psi,p}$. The filter identities verified here establish perfect reconstruction, but do not imply energy preservation. A general construction of dual MRAs is a separate question.

Pair	Low-pass lengths	Zeros at -1	Low-pass phase
Haar / db1 / D2	$2, 2$	$1, 1$	Type II
db2 / D4	$4, 4$	$2, 2$	nonlinear
db4 / D8	$8, 8$	$4, 4$	nonlinear
Biorthogonal 4.4	$9, 7$	$4, 4$	Type I

These examples connect the approximation argument to the filter design. The zeros at -1 determine the number of vanishing moments and hence the cancellation available in the coefficient estimate. The remaining zeros provide choices of phase and symmetry. Splitting reciprocal pairs gives the orthogonal Daubechies factors, while keeping suitable pairs together gives the symmetric biorthogonal factors. In each case, the product-filter identity ensures perfect reconstruction with the stated delay.